

Emergent block structure during training

How a Haar-random `QFTBasis(8,8)` factorises itself into a block code — and why the block **scale** is initial-value dependent

DIV2K-8q, Haar seed-0 · Generated 2026-06-29

Claim

The trained full-image QFT factors itself into a **block code with no block prior** (paper sec5.3): roughly half of the Hadamard-role gates collapse to non-mixing Pauli-Z/X (classical block indices) while the controlled-phase gates stay rich, so the 256×256 transform becomes block-diagonal up to a permutation. The companion endpoint analysis reads this off the **converged** operator. Here we read it off the **training trajectory**: we train one `QFTBasis(8,8)` from Haar-random init (seed 0) and snapshot the live operator at **every** one of the 1008 steps, so we can watch the block code crystallise — and see that **which** block scale it lands on depends on the random start.

Frequency space — the emergence spectrogram

Each snapshot’s operator is read through its mean test-set **coefficient power spectrum**. The 2-D transform is separable, $F = W_t X W_t^\top$, so it factorises per axis; writing $W_t \in \mathbb{C}^{N \times N}$ ($N = 256$) for the 1-D factor and Σ for the image-axis pixel covariance,

$$p_t(f) = [W_t \Sigma W_t^\dagger]_{ff} = \sum_{k,l} W_t[f,k] \Sigma[k,l] \overline{W_t[f,l]} = \mathbb{E}_x \left[\left| (W_t x)_f \right|^2 \right], \quad (1)$$

i.e. the **energy** — the squared coefficient magnitude $\left| (W_t x)_f \right|^2$ — placed in coefficient f , averaged over image lines x and the untransformed axis. Since W_t is unitary up to scale, $\sum_f p_t(f) = \|x\|^2$ (Parseval), so p_t is purely how a fixed energy budget is distributed across coefficients. The covariance is estimated from the 50 DIV2K test images $X^{(n)}$, and the two separable axes are averaged:

$$\Sigma_{\text{row}[k,l]} = \frac{1}{n_{\text{test}} N} \sum_n \sum_j X_{kj}^{(n)} \overline{X_{lj}^{(n)}}, \quad p_t = \frac{1}{2} (p_t^{\text{row}} + p_t^{\text{col}}). \quad (2)$$

The spectrogram colours $\log_{10}(p_t(f) / \max_f p_t(f))$ (peak-normalised per column). The factor W_t is materialised from the checkpoint gates by $W_t[\cdot, k] = T_\theta(e_k)$, and Equation 1 is checked against a direct transform of the test images (agreement $\approx 10^{-13}$).

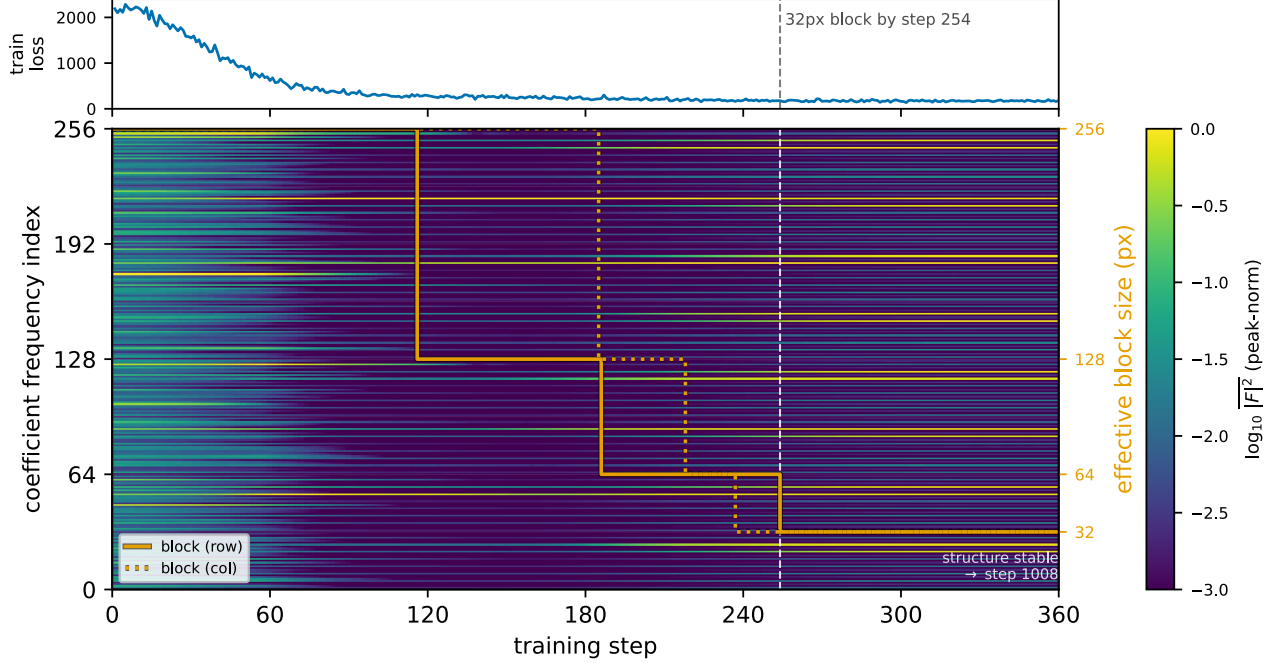


Figure 1: **Training-time Fourier power-spectrum spectrogram.** Mean test-set coefficient power spectrum $p_t(f)$ (peak-normalised $\log_{10}$, viridis) vs training step (linear x , cropped to the emergence window — the structure is flat through the final step 1008); the strip above is the run’s per-step training loss on a shared step axis. At init the Haar-random operator gives a **structureless** spectrum (left) — a near-flat band with a few scattered peaks, not a low-frequency lobe. Over training it reorganises into the block-periodic comb (right). The orange staircase (right axis) is the gate-based effective block size $2^{n_{\text{mix}}}$ for the row (solid) and column (dotted) dimensions, falling $256 \rightarrow 32$ as Hadamard-role gates freeze; the dashed marker is step 254, where both dimensions reach their final 32×32 block.

Two readings stand out. First, the block code forms by a **halving cascade**. The order parameter (orange staircase) is gate-based: for a Hadamard-role gate $U \in \mathbb{C}^{2 \times 2}$ the mixing score is $s = 2|U_{00}||U_{01}| \in [0, 1]$ (1 = Hadamard, 0 = frozen Pauli), and the per-dimension effective block size

$$b = 2^{n_{\text{mix}}}, \quad n_{\text{mix}} = |\{\text{H-role gates with } s > 0.5\}|. \quad (3)$$

Hadamard-role gates freeze one at a time, so b drops $256 \rightarrow 128 \rightarrow 64 \rightarrow 32$ and the spectral comb sharpens in lockstep. The comb is exactly the frequency-space signature of block-diagonality: with the same intra-block transform W_b acting in all N/b blocks, $W_t \approx I_{N/b} \otimes W_b$ up to a permutation, so $p_t(f)$ becomes **periodic in the coefficient index f** with period the block size b — here 32 pixels, i.e. 8 identical sub-spectra (the 8 teeth). Gate collapse and this periodic comb are the same event seen two ways. Second, the cascade **finishes after the loss has largely flattened**: the last halving lands at step 254, well inside the flat top- k MSE valley, so the fine block structure is a property of the optimum rather than something that drives the loss down — consistent with the flat-valley / discrete-basin picture in the seed-robustness study.

Operator space — the same cascade in $|W|$

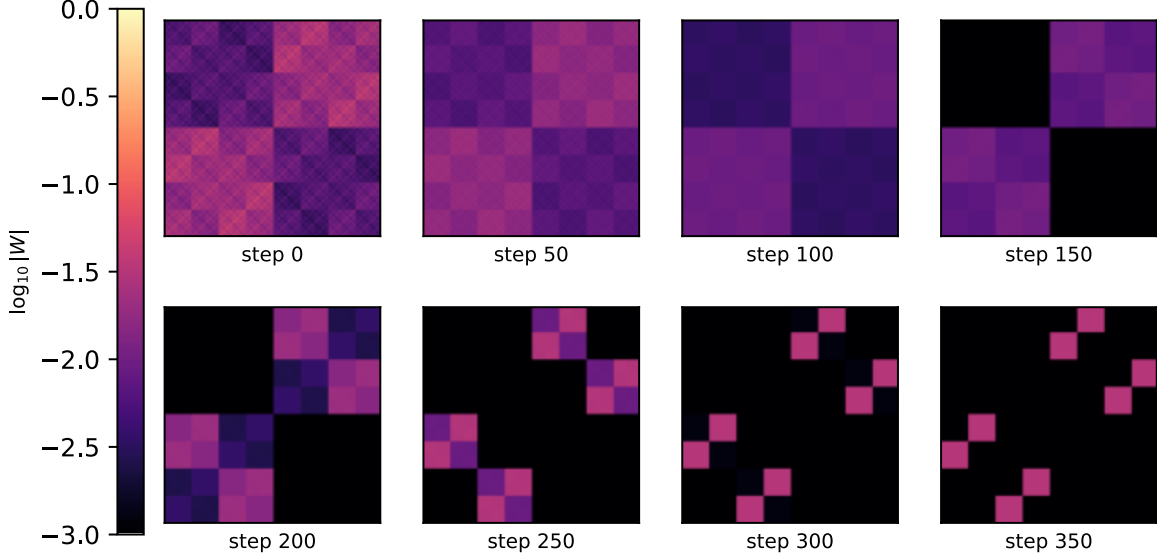


Figure 2: The operator’s 1-D factor $\log_{10}|W|$ at eight training steps. The Haar init (step 0) is dense — every input pixel drives every output coefficient. As gates freeze, $|W|$ collapses onto a block-diagonal (up to the QFT bit-reversal + frozen-X permutation), ending at 8 blocks of 32 pixels. This is the operator-space twin of the frequency-space spectrogram above: the off-block energy draining away **is** the comb sharpening.

How close is it to an exact block transform?

The cleanest way to put a number on “how block” the operator is, is its distance to the nearest exact block transform. For a candidate block size b we project the trained 1-D factor W ($N \times N$, $N = 256$) onto block-diagonal form and keep only what is left over:

1. partition the N indices into $K = N/b$ contiguous blocks;
2. the QFT permutes blocks (bit-reversal + frozen-X), so match each input block to its output block by the Hungarian assignment on the block-energy matrix $C_{ij} = \sum_{\substack{a \in \text{out-block } i, \\ c \in \text{in-block } j}} |W_{ac}|^2$;
3. **project** — zero every entry outside a matched (in $\rightarrow$ out) block — to get $\Pi_b[W]$;
4. **residual** $r(b) = \|W - \Pi_b[W]\|_F / \|W\|_F$: the fraction of the operator’s amplitude lying off the block structure. (It equals $\sqrt{\text{leakage}}$ — the amplitude-domain twin of the off-block **energy** used in the sweep above.)

The full image transform is separable, $T = W_{\text{row}} \otimes W_{\text{col}}$, so its residual is $r_{2D} = \sqrt{1 - (1 - r_{\text{row}}^2)(1 - r_{\text{col}}^2)}$. Sweeping b , r sits near 1 while b is below the true block and then **collapses** — the knee locates the block scale, and the value there is the approximation error.

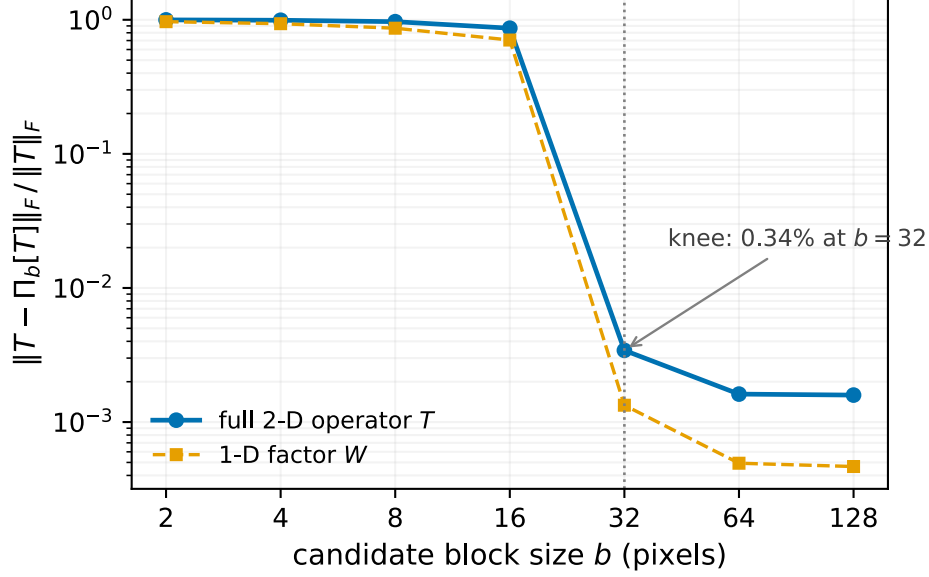


Figure 3: Relative Frobenius distance from the trained operator to its block-diagonal projection, vs candidate block size. A razor-sharp knee at $b = 32$ pixels: the 2-D residual falls from 87% at $b = 16$ to 0.34% at $b = 32$. The trained image transform is thus within 0.34% (relative Frobenius) of an **exact** 32×32 block transform, and is not close to any finer one.

A caveat on the **form** of the block code: the operator is block-**localised** (each block maps to one block, 0.34% off-block) but the intra-block transforms are **not identical** across blocks — they agree only up to per-block diagonal twiddle phases, so the distance to a literally repeated $I_K \otimes W_b$ stays large. This is why the power-spectrum comb still tiles (phase-blind, same per-block power) even though the complex blocks differ.

Which Hadamards freeze

The block code is set by a specific handful of gates. Each H-role gate is classified by its mixing score $s = 2|a||b|$ (Hadamard if $s > 0.5$, else frozen to Pauli-Z when $|a| \geq |b|$ or Pauli-X otherwise). The freezing is **positional**: it is always the most-significant qubits per dimension that collapse to a Pauli — classical block-index bits — while the low qubits keep mixing as the intra-block transform.

This run (seed 0, Figure 4 top): all 16 gates start Hadamard; 6 freeze (2 Pauli-Z, 4 Pauli-X), namely q5– q7 in both dimensions, leaving the 32-pixel block. **The canonical seed** (seed 98, Figure 4 bottom): one more qubit per dimension freezes — q4– q7 — so 8 go Pauli (6 Pauli-Z, 2 Pauli-X), leaving the finer 16-pixel block. Same mechanism one rung deeper: the **number** of frozen qubits, hence the block scale, is what the initialisation selects.

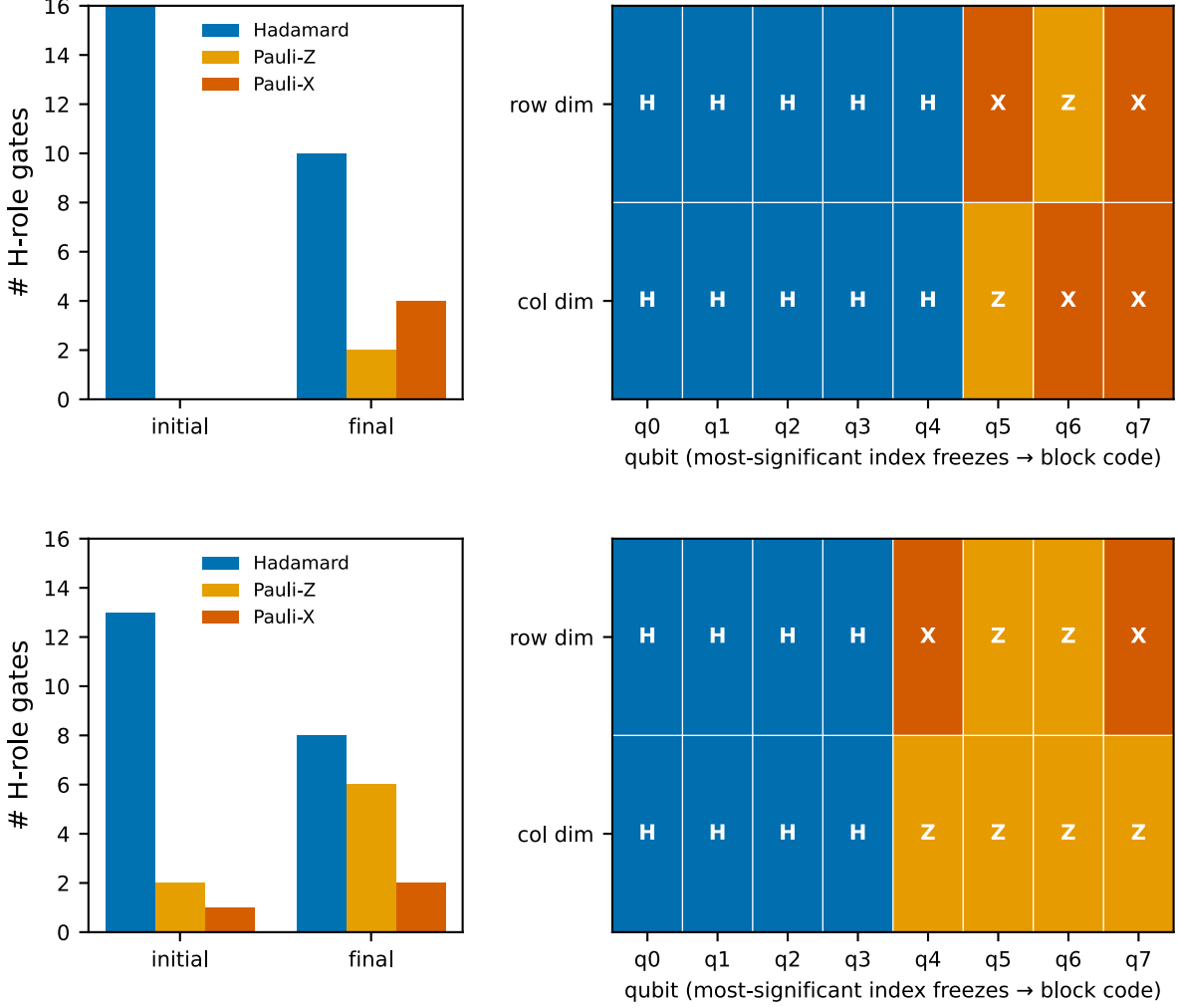


Figure 4: Gate freezing for two initialisations. **Top:** seed 0 — 6 gates freeze (q5–q7 per dimension) into the 32-px block. **Bottom:** seed 98, the canonical paper operator — 8 freeze (q4–q7) into the finer 16-px block. Left panels: H / Pauli-Z / Pauli-X counts, initial vs final; right panels: the exact frozen qubit positions. The most-significant qubits always collapse to Pauli; the initialisation sets how many.

Initial-value dependence

The **existence** of a block factorisation is robust — every Haar start we have trained discovers one with no block prior. The **scale** is not: it is set by how many Hadamard-role gates happen to freeze, which depends on the initialisation. This seed-0 run keeps 5 Hadamards mixing per dimension (freezes 3), leaving a 32-px block (8 blocks). The canonical published operator — a **different** Haar init — freezes one more per dimension (4 vs 3), landing at the finer 16-px block (16 blocks). Same mechanism, different depth: the network reliably finds a block code, while the random start selects which rung of the $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$ cascade it stops on — the operator-level analogue of the seed-to-basin endpoint spread documented in the seed-robustness writeup.